

Reflection - Wasi Imrose

This project allowed me to delve deep into two separate fields that I had no prerequisites for. First the subject matter of Big Data itself and second the complex field of high frequency trading. I initially had difficulty understanding Apache Kafka on a conceptual level. However, the sheer number of iterations I had to make in running the experiments gave me the hands-on learning I was hoping to get. The field of High Frequency Trading was extremely new to me. In fact, this was not my initial choice of topic. However, since I was already working on a forecasting-related project for another class, I decided to take up the challenge of exploring something new. Over the past few months, I picked up a book on Jim Simons on Renaissance Technologies and immersed myself in the field of quantitative trading which served as the philosophical foundation of my experiments.

One of the biggest challenges for this project initially was coordinating with my partner due to his other responsibilities. After a string of communication challenges, we were able to resolve it by dividing the work equally and selecting experiments that were distinct enough, yet aligned with the core topic of using Apache Kafka for HFT. As I was designing my experiments, I deeply researched what software/programs I will be utilizing. My goal was to have the entire pipeline, including the stock simulator, producer, apache, and consumer on the cloud. But the limitation came with connecting the different elements of the pipeline cleanly on the cloud. A jupyter notebook doesn't easily connect to Confluent and vice versa. I tried AWS and the pipeline was extremely laggy when connecting the Python environment with Confluent. Then ultimately I decided to run the producer and consumer scripts on my local computer and use Confluent Kafka as the cloud portion of the pipeline. This still worked because the stock data could come from anywhere, including a locally generated simulator, as long as the ETL pipeline would mimic a cloud based HFT infrastructure. Just off my personal experience in trading, I came up with hypothetical market conditions that would be used to stress test the pipeline. I built the initial baseline simulator using my own Python knowledge and used Claude as a means to guide me through the process and debug any mistakes that I would miss.

The most exciting moment for me was when the pipeline actually worked. I had both the producer and consumer terminals running side by side and I could see the messages sent and received in real time while the metrics were being logged. I then analyzed each metric and tried to understand how it impacted latencies. The process took hours of trial and error and iterations to try new ideas. I don't come from a computer science background with little coding experience in my undergraduate years. So having one of my first ever projects to work seemed like a euphoric moment.

As I reflect on this project and the rest of the class, I am able to directly connect the ideas to my own field. As I am a Mechanical Engineer working in natural gas, my job primarily involves

modeling and simulation of piping networks for day-to-day operations of the gas system serving Upstate NY. This involves analyzing real time SCADA (Supervisory Control and Data Acquisition) data related to gas pressure, velocity and other properties coming directly from the field. I constantly kept thinking how Kafka and cloud infrastructure can easily be integrated into this data pipeline in a reliable way. I was able to further build on this during the data center field trip. I have worked in HVAC controls in the past and had already known the mechanical side of the cooling system. But through this class, I was also able to connect the software and data aspect of it as well. What tied it all together were the in-depth ethical discussion we had in class on the real cost of AI and datacenter and its repercussions on a global scale. So much so that I brought up these ideas in my work meetings to further the discussion. Overall, I thoroughly learned the nuts and bolts of big data with further goals to delve deeper. The combination of big data and data-in-movement will stick with me as I continue developing in my field!